12. Rigidity and boundary modes of periodic Maxwell frames

Pure Thought research portfolio

Revised 17 September 2026

12. Rigidity and boundary modes of periodic Maxwell frames

Revised specification — 2026-09-11. Detailed critique · Original description · Evidence standard

Plain-language guide

The problem in everyday terms

A framework of bars or springs can be rigid in one place and easy to deform in another. Sometimes a carefully arranged repeating framework has soft motions concentrated at its boundary. The question is whether we can predict and prove those motions from the exact geometry, rather than simply observing them in a simulation. Counting connections helps, but the direction of each connection matters too.

Key terms

- **Frame** — A network of sites joined by bars or springs, used as an ideal mechanical model.
- **Degree of freedom** — An independently variable coordinate of the system's motion.
- **Isostatic or Maxwell frame** — A frame balanced between degrees of freedom and constraints in the relevant counting sense; details of zero modes and self stress still matter.
- **Zero mode** — An infinitesimal displacement requiring no restoring energy in the linearized model.
- **State of self stress** — A pattern of internal bond tensions that balances forces without external loads.
- **Compatibility matrix C** — The linear map from small site displacements to bond extensions.
- **Stiffness matrix K** — The operator relating displacements to restoring forces in the linear approximation.
- **Winding number** — An integer counting how a complex quantity winds around zero along a closed path; here it helps characterize the periodic frame.
- **Termination** — The precise way a periodic structure is cut to make a boundary.

Why this matters

Geometry-controlled softness could help explain and design structures with directional compliance, boundary motion or localized mechanical response. The scientific value is understanding which behaviors follow from a robust index and which are accidental features of one geometry. That distinction matters before translating an ideal spring model into a physical structure.

What progress would mean

A useful result would give a frame, an exact count of its allowed soft motions and a proof of where they concentrate. Material failure, nonlinear deformation and manufacturing tolerance would remain additional engineering questions.

Core question

For a geometrically specified isostatic frame and boundary termination, can the zero-mode count and localization be proved from its compatibility matrix?

Scope and assumptions

Use unstressed central-force springs with exact rational or algebraic site coordinates, positive spring constants, and explicit periodic bond offsets. Fix the unit-cell gauge, dimension, masses, and boundary termination. Begin with a small frame and a known periodic Maxwell lattice.

Connectivity alone is insufficient: bond directions determine the compatibility matrix. Distinguish rigid-body motions, floppy modes, and states of self stress. Claims concern linearized ideal mechanics.

Mathematical target

Let C map displacements to bond extensions and Q=C†. For a finite free frame,

$$n_0 - n_{ss} = dN - N_b, \qquad K = C^\dagger \operatorname{diag}(k_b) C,$$

with dynamical matrix $\mathrm{M}^{-1/2}\mathrm{K}\mathrm{M}^{-1/2}$. For a square periodic C(k), define winding $(2\pi\mathrm{i})^{-1}\oint \mathrm{d} \log \det \mathrm{C}(\mathrm{k})$ only along loops where the determinant is nonzero. Account for acoustic translation zeros and the local boundary count when translating bulk winding into an edge index.

Required outputs and proof obligations

Export positions, bonds, C, exact rank/kernel calculations, self-stress basis, winding certificate, and a boundary-specific mode count. For localization, provide a transfer-matrix/root-modulus bound separated from the unit circle. Specify perturbations preserving the nonvanishing condition and positivity of spring constants.

Validation and rejection controls

Check one elementary underconstrained and one overconstrained frame against the index equation. Verify K is positive semidefinite and translations are zero modes where appropriate. Compare two terminations of the same bulk. Reject winding calculations crossing a zero of det C.

Milestones and research extension

Baseline: exact Maxwell–Calladine counts on finite frames. Strong: one periodic frame with certified winding and termination-dependent edge localization. Research extension: a bounded family with robust mode counts. Elastic beams, prestress, nonlinear stability, and manufacturing tolerance require separate models.

Milestones are acceptance gates, not calendar promises. Before a run, freeze the concrete instance/domain, tolerance, compute budget, and permitted dependencies. A bounded search that fails to find a result must preserve an unresolved outcome.

Starting instruction

Work on the baseline above. First make every input and convention explicit, then validate the reference case. Build the checker alongside the calculation. Label numerical evidence, finite-model certificates, and continuum theorems separately; return the strongest justified result and identify the exact unmet proof obligation.

Primary references and their role

- Kane and Lubensky, Topological Boundary Modes in Isostatic Lattices — Compatibility matrices, topological indices, and boundary modes.

References support definitions or limitations; the scope choices and proposed milestones are editorial recommendations, not claims that these results have already been obtained.

Displayed formulation and claim boundary

Canonical specification.

$$Q = C^{\dagger}, \qquad n_0 - n_{ss} = dN - N_b, \qquad K = C^{\dagger}\,\mathrm{diag}(k_b)C, \quad D = M^{-1/2}KM^{-1/2}.$$

$$\nu_\gamma = \frac{1}{2\pi i}\oint_\gamma d\log\det C(k), \qquad \det C(k) \neq 0 \text{ on } \gamma.$$

The first count includes rigid-body motions and self stress. Supply actual bond directions, unit-cell gauge and termination. Acoustic zeros and the local boundary count must be included before identifying edge modes from winding.

Detailed review and rationale

Revised problem · Preserved original · Pinned upstream source

Line references below refer to the preserved original at commit `f854d4e12b55`, not the rewritten file. "Blocking" means the original issue can invalidate the scientific target or claimed certificate; "major" means it materially changes scope, inference, or acceptance criteria. These are critiques of specifications, not results of running the embedded research-code sketches.

Detailed findings

1. **Blocking — counting formula (22–27, 100).** The original replaces the Maxwell–Calladine difference between zero modes and self stresses with an incorrect count. It can mislabel even simple rigid structures.
2. **Blocking — mechanical operator (75–78).** The displayed dynamical matrix does not provide a reliable central-force construction. Building C from bond geometry and K=C†kC supplies a checkable definition.
3. **Major — graph-only topology (38, 61–64).** Frames with the same graph can have different rigidity and polarization. Geometric embedding and unit-cell conventions are part of the input.

4. **Major — boundary and stability claims (87–103).** A Berry phase alone does not give a universal zero-mode count, and a passive unstressed stiffness matrix does not have generic negative-frequency-squared spectral flow. Boundary indices and transfer decay replace the unsupported shortcut.

What the revision preserves and changes

The central motivation is retained, but the executable question is now: **For a geometrically specified isostatic frame and boundary termination, can the zero-mode count and localization be proved from its compatibility matrix?** The revised scope chooses a concrete starting model; those choices are proposed research-design decisions, not facts inferred from the original. The baseline can succeed without resolving the full open research program.

The replacement also removes the original implementation sketches from the active specification. They remain in the archive for comparison, but have not been made executable or certified. New implementations should derive their equations from the revised definitions rather than copy unchecked prototypes.

Acceptance criteria to use instead

Baseline: exact Maxwell–Calladine counts on finite frames. Strong: one periodic frame with certified winding and termination-dependent edge localization. Research extension: a bounded family with robust mode counts. Elastic beams, prestress, nonlinear stability, and manufacturing tolerance require separate models.

Sources supporting the corrections

- Kane and Lubensky, Topological Boundary Modes in Isostatic Lattices — Compatibility matrices, topological indices, and boundary modes.

Evidence and execution standard

This standard applies to the revised specifications. It is a proposed contract for future research runs; this editorial revision does not supply the scientific certificates themselves.

What "pure thought" means here

Use explicitly stated mathematical models and deductions without fitting conclusions to experimental outcomes. Synthetic examples, exact model inputs, analytic references, numerical search, and established software are allowed. Physical constants, electronic-structure approximations, constitutive laws, and empirical parameter sets must be identified when used. A model-derived result is conditional on that model.

The original ban on legacy software conflicts with its proposed reliance on NumPy, SciPy, CVXPY, nauty, and other libraries. Reimplementing every optimizer and algebra system is not necessary for scientific independence. Prefer a small independently checkable certificate and a documented trusted base. A fresh implementation can be a project objective, but must not be confused with a mathematical proof or novelty claim.

Four evidence levels

Label	What it establishes	What it does not establish
Numerical evidence	A reproducible computation at stated precision, sample size, and discretization	Exact truth, continuum coverage, or a universal theorem
Finite-model certificate	An exact/validated statement about the specified finite model or finite relaxation	An omitted tail, thermodynamic limit, or physical realization
Conditional mathematical theorem	A proof over the stated domain with every hypothesis and error controlled	Truth outside those assumptions or a demonstrated physical system
Formal proof	A checked theorem in a named proof assistant, with imported axioms/dependencies listed	Correct physical modeling merely by virtue of formalization

Formalization is an optional extra deliverable unless the specific project makes it mandatory. High precision, JSON output, solver status and small residuals do not by themselves confer any of these proof levels.

Claim directions

For minimization over a larger feasible set, the optimum is a lower bound on the original minimum. A physical feasible point supplies an upper bound. For maximization the directions reverse. Track feasibility of the primal/dual witnesses separately from their objective values.

Infeasibility of an inner ansatz excludes only that ansatz. Infeasibility of an outer relaxation can exclude the original problem if the map into the relaxation is proved and all approximations preserve necessity. Passing necessary tests means "not excluded"; existence needs a constructive sufficient argument.

A smallest observed example is not a minimality theorem. A complete catalog must define the universe, equivalence relation, boundaries, and proof that pruning misses nothing. A finite-size numerical trend does not prove an asymptotic limit.

Minimum run manifest

Before calculation, record the problem ID and revision hash, exact inputs and conventions, quantified domain, equivalence relation, target observable, accepted evidence level, tolerances, and resource limits. Record dependencies and precision/rounding mode. Where randomization is used, preserve seeds, trial counts and raw sufficient statistics. Resource limits are planning decisions, not evidence that a mathematical solution cannot exist.

An output bundle should include:

- The statement being claimed, including all assumptions and quantifiers.
- Construction or proof object, plus an independently runnable verifier and its trust assumptions.

- Numerical/error ledger separating roundoff, discretization, truncation, optimization, sampling and modeling error where relevant.
- Positive reference cases and negative controls that the verifier must reject.
- Coverage: included domain, omitted domain, unresolved regions and termination reason.

These are requirements for future implementations, not a promise of polynomial-size certificates for every problem.

Checking specific certificate types

LP/SDP/SOS: verify equality constraints, objective direction, and cone feasibility using exact arithmetic or outward-rounded intervals. Rationalizing a numerical solution can break PSD constraints. Correct residuals with justified norm bounds; do not silently project and reuse the old objective. Distinguish a proof of a polynomial approximation from a proof for the original function.

Interval numerics: state function enclosures, domains and outward rounding. A small root box needs existence and, if claimed, uniqueness. Cover the complement for a global enumeration. Include singularities and tails rather than discarding them numerically.

SAT/SMT: a solver's UNSAT label is not a certificate. Preserve a proof supported by a checker and the proof that the encoding, symmetry reduction and blocked solutions match the mathematical universe. DRAT applies to an appropriate Boolean CNF proof, not automatically to every real-arithmetic or SDP claim.

Statistics: specify the estimand, dependence assumptions, confidence procedure and bias. Zero observed failures does not prove zero failure probability. MCMC error bars based on estimated autocorrelation are not generally rigorous mixing guarantees.

Milestones and novelty

Baseline means a working, independently validated result on a concrete reference problem. Strong means a broader or sharper result with its claimed evidence level. Research extension means a possible novel result whose feasibility and significance need review. A negative theorem can be valuable; an unsuccessful search can also be useful if its scope is documented, but the two are not interchangeable.

Do not promise "publication-quality" in a fixed number of months. Before claiming novelty, conduct a focused current literature comparison against the exact theorem/model/metric. The sources in this revision check key definitions and limitations; they are not an exhaustive novelty audit across all 30 fields.